

Research Methodology in CS (CSEG3060_3)

Lecture 3: Developing a Research Proposal: identifying research problems, formulating objectives and hypotheses, literature review, significance, scope, and feasibility of the study

Unit 4: Technical Writing and Research Proposal Development • Mapped Course Outcome: CO4

Lecture 3: Developing a Research Proposal

Course: Research Methodology in Computer Science (CSEG3060_3) **Program:** B.Tech (Computer Science Engineering) **Semester:** V **Unit:** 4 — Technical Writing and Research Proposal Development **Mapped Course Outcome:** CO4 **Estimated Duration:** 60 minutes

1. Learning Objectives

By the end of this lecture, students should be able to:

1. Distinguish between a research problem, a research question, and a research hypothesis.
 2. Apply structured techniques to identify, refine, and justify a research problem in computer science.
 3. Formulate clear, measurable research objectives and testable hypotheses.
 4. Conduct and synthesize a focused literature review aligned with a chosen problem.
 5. Articulate the significance, scope, and feasibility of a proposed study with rigor.
 6. Critically evaluate a sample research proposal against standard academic criteria.
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2. Introduction

A **research proposal** is a formal document that outlines the intended research, justifies its necessity, and describes the methods by which the investigator will conduct it. In computer science, where projects often involve rapidly evolving technologies and complex systems, the proposal serves as a roadmap for the researcher and a contract of academic accountability with reviewers, funding agencies, and supervisors.

This lecture dissects the *intellectual core* of a research proposal: the formulation of the problem, the construction of objectives and hypotheses, the role of the literature review, and the assessment of significance, scope, and feasibility. Mastery of these elements ensures that

subsequent technical writing and presentation efforts are grounded in a defensible scientific rationale.

3. Key Concepts Overview

Element	Purpose
Research Problem	Defines the gap or issue the study addresses
Research Questions	Operationalizes the problem into answerable inquiries
Objectives	States the concrete aims of the investigation
Hypotheses	Provides testable predictions (where applicable)
Literature Review	Establishes context, identifies gaps, and avoids duplication
Significance	Justifies why the research matters
Scope	Delimits what the study will and will not cover
Feasibility	Demonstrates that the study can realistically be executed

4. Identifying the Research Problem

4.1 What Constitutes a Research Problem?

A **research problem** is a specific issue, difficulty, or contradiction that can be investigated through empirical, theoretical, or computational means. In computer science, problems typically arise from:

- **Performance bottlenecks** in existing algorithms or systems.
- **Limitations** of current models in handling new data types or scales.
- **Unmet user requirements** in human–computer interaction contexts.
- **Open theoretical questions** in formal models (e.g., complexity theory, computability).
- **Sociotechnical concerns** such as privacy, fairness, or sustainability of digital systems.

A well-stated problem answers three implicit questions:

1. *What* is the observed phenomenon or deficiency?
2. *Where* and *under what conditions* does it occur?
3. *Why* does it matter to the field, stakeholders, or society?

4.2 Sources of Research Problems

- **Personal experience** with a system, tool, or framework.
- **Literature gaps** identified through surveys and systematic mapping studies.
- **Industry reports** (e.g., Gartner, ACM TechReports) that highlight unsolved challenges.
- **Conference proceedings** (NeurIPS, SIGCOMM, CHI) that propose open problems.
- **Replications and contradictions** in prior empirical findings.
- **Interdisciplinary intersections**, such as applying NLP to legal corpora or using blockchain for healthcare records.

4.3 Techniques for Problem Identification

1. **Brainstorming and Mind Mapping.** Generate broad themes and cluster them into narrower sub-issues.
2. **The 5W1H Framework.** Apply *What, Why, Who, Where, When, How* to dissect a broad topic.
3. **PICO(T) Method (adapted from evidence-based research).**
 - **Population:** the data or system users.
 - **Intervention:** the proposed technique or model.
 - **Comparison:** baseline or state-of-the-art.
 - **Outcome:** measured metric (accuracy, latency, energy).
 - **Timeframe:** evaluation period.
4. **Problem Tree Analysis.** Map causes, effects, and the core issue.
5. **Critical Reading.** Identify limitations explicitly mentioned by authors of seminal papers.

4.4 Characteristics of a Good Research Problem

A good problem statement is **Specific, Measurable, Achievable, Relevant, and Time-bound (SMART)**:

- *Specific:* Focused on a defined sub-domain (e.g., "intrusion detection in IoT networks using deep learning" rather than "cybersecurity").
- *Measurable:* Outcomes can be quantified (precision, recall, F1, throughput).
- *Achievable:* Within the technical and resource capacity of the researcher.
- *Relevant:* Addresses a recognized gap or stakeholder need.
- *Time-bound:* Can be completed within the project duration.

4.5 Example from Computer Science

Weak statement: "Machine learning is not very accurate." **Strong statement:** "Existing convolutional neural network architectures for skin lesion classification achieve a maximum F1-score of 0.81 on the ISIC 2019 dataset, which is insufficient for clinical deployment where false negatives carry high patient risk. This study seeks to improve classification accuracy while maintaining interpretability."

The latter statement clearly identifies the *what*, *why*, and *so what*.

5. Formulating Research Objectives

5.1 Purpose of Objectives

Objectives translate the problem into **actionable intents**. They describe what the researcher intends to accomplish and serve as benchmarks for evaluating the project's success.

5.2 Classification of Objectives

- **Primary (General) Objective:** The overarching goal that captures the spirit of the research.
- **Secondary (Specific) Objectives:** Discrete, sequential sub-goals that contribute to the primary objective.

5.3 Characteristics of Well-Formed Objectives

Effective objectives are:

1. **Clear** — Unambiguous and written in plain language.
2. **Realistic** — Achievable given constraints.
3. **Measurable** — Allow evaluation against specific metrics.
4. **Ordered** — Sequenced logically (foundational before applied).
5. **Action-oriented** — Begin with strong verbs such as *design*, *develop*, *evaluate*, *compare*, *analyze*, *implement*, *validate*.

Avoid verbs such as *study*, *understand*, or *explore* when they lack a measurable complement.

5.4 Example: Smart Traffic Management System

Primary Objective: To design and evaluate a deep reinforcement learning framework for adaptive traffic signal control in heterogeneous urban environments.

Secondary Objectives:

1. To analyze the limitations of existing fixed-time and actuated traffic control systems through a systematic literature review.
2. To develop a multi-agent reinforcement learning model that dynamically adjusts signal phases based on real-time traffic density.
3. To implement the proposed framework in a simulated urban grid using SUMO and to benchmark it against baseline controllers.
4. To evaluate performance across metrics of average vehicle delay, throughput, queue length, and emissions.
5. To assess the transferability of the trained policy across varying traffic demand profiles.

6. Formulating Hypotheses

6.1 What is a Hypothesis?

A **hypothesis** is a tentative, testable prediction about the relationship between two or more variables. In computer science, hypotheses are particularly relevant for experimental, empirical, and analytical research where causal or correlational claims are made.

6.2 Types of Hypotheses

Type	Description	Example
Null Hypothesis (H_0)	States that there is no significant effect or difference	"There is no significant difference in classification accuracy between the proposed CNN model and the baseline ResNet-50 on the CIFAR-10 dataset."
Alternative Hypothesis (H_1 or H_a)	Asserts a meaningful effect or difference	"The proposed CNN model achieves significantly higher classification accuracy than ResNet-50 on CIFAR-10."
Directional Hypothesis	Specifies the expected direction of the effect	"Increasing the number of transformer layers beyond 12 will degrade performance on low-resource translation tasks due to overfitting."
Non-directional Hypothesis	Specifies an effect without direction	"There is a relationship between dataset size and model robustness."

6.3 Characteristics of a Good Hypothesis

- **Testable:** Can be validated using statistical or empirical methods.
- **Falsifiable:** Can, in principle, be disproved.
- **Specific:** Clearly identifies variables and their relationship.
- **Grounded:** Derived from theory or prior literature.
- **Concise:** Ideally expressible in a single declarative sentence.

6.4 Hypothesis Formulation: Mathematical Expression

A general form for a comparative hypothesis in performance evaluation can be expressed as:

$$H_0: \mu_{\text{proposed}} - \mu_{\text{baseline}} = 0 \quad H_1: \mu_{\text{proposed}} - \mu_{\text{baseline}} \neq 0$$

where μ represents the population mean of a performance metric (e.g., accuracy, latency, throughput). The researcher collects sample data and conducts a statistical test (e.g., paired t-test, Wilcoxon signed-rank test) to determine whether to reject H_0 at a chosen significance level α (typically 0.05).

6.5 When Hypotheses Are Not Applicable

In **exploratory, descriptive, or design-science research**, hypotheses may be inappropriate. For example, a study that proposes a new software architecture pattern does not always require hypothesis testing; instead, it may employ **design principles, heuristics, or qualitative evaluation criteria** such as utility, elegance, and reusability.

7. Literature Review

7.1 Purpose of the Literature Review

The literature review serves four interrelated functions in a research proposal:

1. **Establishes the scholarly context** of the proposed work.
2. **Identifies gaps** that justify the new study.
3. **Prevents duplication** of prior efforts.
4. **Informs methodological choices** by surfacing effective techniques, datasets, and evaluation practices.

7.2 Types of Literature Review

Type	Description	Use Case
Narrative	Broad synthesis of the field	Doctoral proposals, introductions
Systematic	Protocol-driven search, inclusion criteria, quality assessment	Evidence-based CS research
Mapping Study	Categorizes the landscape of research	Software engineering, HCI
Meta-analysis	Statistical aggregation of effect sizes	Algorithm benchmarking
Scoping Review	Maps breadth and depth of literature	Emerging topics (e.g., quantum ML)

7.3 Conducting a Systematic Literature Review (SLR)

The standard SLR process follows the **Kitchenham guidelines**:

1. **Define research questions** for the review.
2. **Develop a search strategy** (databases: ACM DL, IEEE Xplore, Scopus, arXiv; keywords; inclusion/exclusion criteria).
3. **Select primary studies** through multi-stage screening.
4. **Assess study quality** using validated checklists.
5. **Extract data** using standardized forms.
6. **Synthesize findings** thematically or quantitatively.

7.4 Structuring the Literature Review Section

A well-structured review progresses logically:

- **Background:** Define key concepts and their evolution.
- **Theoretical Foundations:** Anchor the work in established theories (e.g., PAC learning, distributed systems theory).
- **State of the Art:** Summarize contemporary approaches and their performance.
- **Critique:** Identify limitations, contradictions, and methodological weaknesses.
- **Gap Identification:** Explicitly state what is missing and how the proposed work addresses it.

7.5 Avoiding Common Pitfalls

- **Annotation without synthesis:** Listing paper summaries rather than integrating insights.
- **Citation bias:** Over-representing Western, English-language, or industry sources.
- **Outdated evidence:** Ignoring recent preprints or open-source benchmarks.
- **Lack of critical voice:** Accepting claims at face value without evaluating methodology.

7.6 Mini Case Study: Federated Learning for Healthcare

A literature review on "**Privacy-Preserving Federated Learning for Electronic Health Records**" might trace progression from centralized machine learning approaches (Konečný et al., 2016) to differentially private federated systems (Wei et al., 2020), to recent advances in personalization under non-IID data (Li et al., 2023). The synthesis would highlight recurring limitations such as communication overhead, heterogeneity, and adversarial vulnerability, thus justifying a new contribution that introduces, for example, a lightweight secure aggregation protocol.

8. Significance of the Study

8.1 Defining Significance

Significance conveys the **contribution** of the proposed research to theory, practice, or society. Reviewers and funding agencies expect a crisp articulation of why the study matters.

8.2 Dimensions of Significance

1. **Theoretical Contribution:** Advances conceptual understanding (e.g., a new complexity bound or model).
2. **Methodological Contribution:** Introduces improved procedures, tools, or evaluation frameworks.
3. **Empirical Contribution:** Provides new datasets, benchmarks, or experimental insights.
4. **Practical Contribution:** Solves an industry problem, informs policy, or improves user experience.
5. **Societal Contribution:** Addresses ethical, environmental, or equity considerations.

8.3 Writing a Strong Significance Statement

Template:

"This study is significant because it [addresses / resolves / extends] [specific gap] in [domain]. The findings will [advance theory / inform practice / support stakeholders] by [specific mechanism]."

Example:

"This study is significant because it addresses the lack of lightweight intrusion detection models for edge devices in IoT networks. The proposed framework will advance theory in resource-constrained deep learning and will offer practitioners a deployable artifact capable of identifying threats in real time without relying on cloud infrastructure."

9. Scope of the Study

9.1 Purpose of Scope Definition

Scope establishes the **boundaries** of the research. It ensures clarity, manages stakeholder expectations, and prevents scope creep.

9.2 What Scope Encompasses

- **Topic boundaries:** Specific algorithms, systems, or phenomena covered.
- **Theoretical boundaries:** Frameworks adopted or excluded.
- **Methodological boundaries:** Methods, tools, and datasets used.
- **Geographical or demographic boundaries:** Relevant for social computing, HCI, or policy-oriented CS research.
- **Temporal boundaries:** Time period of data collection or system operation.

9.3 Sample Scope Statement

"This study is limited to the evaluation of deep reinforcement learning algorithms (PPO, SAC, DQN) for adaptive traffic signal control in simulated urban grids of up to 20 intersections. Real-world deployment, large-scale metropolitan networks, and pedestrian-priority optimization fall outside the scope of this study."

9.4 Common Scope Errors

- **Too broad:** Attempting to solve multiple unrelated problems.
 - **Too narrow:** Insufficient to yield meaningful conclusions.
 - **Unstated assumptions:** Implicitly assuming data availability or hardware access without verification.
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10. Feasibility of the Study

10.1 Definition

Feasibility demonstrates that the proposed research can be **realistically executed** within the constraints of time, expertise, equipment, data, and budget.

10.2 Pillars of Feasibility Assessment

Dimension	Key Questions
Technical	Are required tools, libraries, and hardware accessible?
Data	Are datasets available, accessible, and legally usable?
Operational	Are institutional approvals (IRB, ethics) in place?
Schedule	Is the timeline realistic given task dependencies?
Economic	Is the budget adequate for computation, travel, and dissemination?
Human Resources	Does the team possess the necessary expertise?

10.3 Feasibility Analysis Techniques

- **SWOT Analysis:** Strengths, Weaknesses, Opportunities, Threats.
- **Gantt Chart:** Visualizes task scheduling and milestones.
- **Risk Register:** Identifies risks, likelihood, impact, and mitigation strategies.
- **Proof-of-Concept (PoC):** A small-scale preliminary implementation to validate assumptions.

10.4 Example: Feasibility Block in a CS Proposal

"The proposed study is technically feasible because all required libraries (PyTorch 2.1, Ray RLLib) are open-source, and computation can be performed on the institutional GPU cluster (4× NVIDIA A100). The PEMS-BAY traffic dataset is publicly available through the California Department of Transportation. A preliminary PoC using SUMO on a 4-intersection grid has been completed, confirming that a single training episode requires approximately 12 minutes, well within the project's computational budget."

10.5 Feasibility Risk Categories in CS Research

1. **Data Risks:** Insufficient or biased datasets, licensing restrictions.
 2. **Compute Risks:** Hardware failures, GPU shortages, cost overruns.
 3. **Algorithmic Risks:** Non-convergence, instability, poor generalization.
 4. **Operational Risks:** Staff attrition, dependency on third-party APIs.
 5. **Ethical Risks:** Use of personal data, dual-use concerns in security research.
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11. Illustrative Case Study

Title: *Hybrid Quantum-Classical Neural Networks for Early Detection of Diabetic Retinopathy*

Research Problem: Existing CNN models for diabetic retinopathy detection require substantial labeled data and computational resources, limiting their deployment in resource-constrained rural clinics.

Research Questions:

1. How does a hybrid quantum-classical neural network compare against a purely classical CNN in classification performance on the Kaggle APTOS 2019 dataset?
2. Can hybrid architectures reduce the number of trainable parameters without sacrificing accuracy?
3. What is the effect of quantum noise and circuit depth on model robustness?

Objectives:

1. To design a variational quantum circuit integrated with a classical feature extractor.
2. To train and evaluate the hybrid model on APTOS 2019 using 5-fold cross-validation.
3. To compare performance with baseline ResNet-50 and EfficientNet-B0.

4. To analyze the trade-off between parameter count and accuracy.

Hypotheses:

$$H_0: \mu_{\text{accuracy}}^{\text{hybrid}} = \mu_{\text{accuracy}}^{\text{ResNet-50}} \quad H_1: \mu_{\text{accuracy}}^{\text{hybrid}} \neq \mu_{\text{accuracy}}^{\text{ResNet-50}}$$

Literature Review Highlights: Synthesizes prior work on quantum kernel methods (Havlicek et al., 2019), variational quantum circuits (Cerezo et al., 2021), and medical image classification benchmarks.

Significance: Contributes to quantum machine learning theory while addressing a tangible healthcare delivery gap.

Scope: Limited to binary classification (referable vs. non-referable); quantum simulation via PennyLane rather than physical quantum hardware.

Feasibility: Datasets are publicly accessible; simulation runs on institutional HPC; preliminary circuit designs have been validated in PennyLane.

12. Summary

This lecture has traversed the foundational components of a research proposal in computer science:

- The **research problem** anchors the inquiry, articulating a precise and significant gap.
- **Research questions** decompose the problem into answerable units; **objectives** operationalize the intent of the researcher; **hypotheses** provide testable predictions where appropriate.
- The **literature review** situates the work in scholarly context, synthesizes prior contributions, and exposes the gap that the proposal will fill.
- The **significance** statement justifies the contribution across theoretical, methodological, practical, and societal dimensions.
- The **scope** clearly delimits the work to maintain focus and manage expectations.
- The **feasibility** assessment validates that the proposed research can be conducted with available resources and within realistic constraints.

Together, these elements form the intellectual scaffold upon which the methodological design, ethical considerations, and project timeline of the proposal will subsequently rest.

13. Practice and Review Questions

13.1 Short-Answer Questions

1. Differentiate between a research problem and a research question with a suitable example from computer science.
2. Why are SMART criteria important when formulating a research problem? Illustrate with an example in software engineering.
3. Distinguish between a null hypothesis and an alternative hypothesis. Under what conditions would you use a directional hypothesis?
4. State two reasons why a literature review is essential before finalizing a research proposal.
5. Differentiate between *scope* and *feasibility* in a research proposal.

13.2 Long-Answer / Analytical Questions

1. Consider a real-world domain (e.g., AI in agriculture, blockchain in supply chains, NLP for low-resource Indian languages). Draft a complete research problem statement, three research objectives, and two testable hypotheses for a proposed study.
2. Critically evaluate the following problem statement: "*This project aims to use AI to solve traffic problems.*" Rewrite it according to SMART criteria and justify each improvement.
3. Conduct a mini systematic literature mapping (at least 10 papers) on a chosen topic in CS. Identify three themes and one unresolved gap. Present the synthesis as a structured paragraph.
4. Imagine a faculty review committee has rejected your proposal citing "lack of feasibility." Outline the specific elements you would revise and justify your reasoning.
5. For a hypothetical federated learning project, prepare a feasibility plan that addresses technical, data, operational, and ethical risks.

13.3 Multiple-Choice Questions

1. Which of the following is the strongest example of a research problem statement?
 - (a) AI is interesting.
 - (b) There is a need for better AI.
 - (c) Current transformer-based chatbots hallucinate in 22% of responses on the LegalQA dataset, exposing users to legal misinformation.
 - (d) AI will change the world.
2. A null hypothesis in a comparative algorithm study typically states that:

- (a) The proposed algorithm is always better.
- (b) There is no significant difference between the algorithms on the chosen metric.
- (c) The proposed algorithm will fail.
- (d) The dataset is irrelevant.

3. Which section of a research proposal justifies *why* the study matters?

- (a) Scope
- (b) Feasibility
- (c) Significance
- (d) Methodology

4. The Kitchenham guidelines are most associated with:

- (a) Writing systematic literature reviews in software engineering.
- (b) Developing neural network architectures.
- (c) Designing software testing frameworks.
- (d) Conducting user interviews in HCI.

5. Scope creep is best prevented by:

- (a) Expanding objectives iteratively.
- (b) Avoiding deadlines.
- (c) Clearly defining boundaries of the study in the proposal.
- (d) Adding more team members.

Answers: 1-(c), 2-(b), 3-(c), 4-(a), 5-(c).

14. Exam Preparation Tips

1. **Memorize the structural anatomy of a proposal.** Examiners frequently ask students to outline the components in correct sequence.
2. **Practice hypothesis writing.** Be prepared to convert objectives into testable H_0 and H_1 formulations.
3. **Distinguish similar terms.** Know the differences between *problem-question-objective-hypothesis*, and between *scope-feasibility-significance*.
4. **Use domain-specific examples.** Citing computer science examples (NLP, IoT, HCI, federated learning) earns more marks than generic statements.

5. **Link every objective to a hypothesis or evaluation metric.** Proposals are scored on internal consistency.
 6. **Be precise with verbs.** Use measurable verbs in objectives (*design, evaluate, benchmark*) rather than vague ones (*study, understand*).
 7. **Anticipate reviewer objections.** Practice preempting concerns related to data availability, compute, and ethics.
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15. Suggested Readings

1. Creswell, J. W., & Creswell, J. D. (2018). *Research Design: Qualitative, Quantitative, and Mixed Methods Approaches* (5th ed.). SAGE Publications.
 2. Kitchenham, B., & Charters, S. (2007). *Guidelines for Performing Systematic Literature Reviews in Software Engineering*. EBSE Technical Report.
 3. Sekaran, U., & Bougie, R. (2020). *Research Methods for Business* (7th ed.). Wiley.
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End of Lecture 3 Notes — Unit 4, Research Methodology in Computer Science (CSEG3060_3).